

MARGARET GATHONI

Full Stack Developer | Data Scientist | Tech Lead

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PROFESSIONAL SUMMARY

Full Stack Developer and Data Science professional with 6+ years of experience building scalable systems, data pipelines, and analytical solutions across healthcare, enterprise, and research sectors. Proven expertise in Python/Django/Flask, React, PostgreSQL, cloud infrastructure (AWS), and machine learning. Track record of leading technical teams, automating workflows that saved 80% of manual effort, and training 200+ users on digital systems. Currently pursuing MSc in Data Science and MBA in Strategic Management.

CORE COMPETENCIES

Programming: Python, JavaScript, C#, SQL **Frameworks:** Django, Flask, FastAPI, React, NextJS
Databases: PostgreSQL, MySQL, Oracle, MongoDB **Cloud & DevOps:** AWS, Docker, Jenkins, CI/CD
Data Science: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch **BI Tools:** Power BI, Tableau, Looker
Game Dev: Unity 3D, C#, WebGL, Mobile Games **Other:** REST APIs, ETL Pipelines, Microservices

PROFESSIONAL EXPERIENCE

HMIS Specialist

CIHEB Kenya | Mar 2025 - Sep 2025 | Nairobi, Kenya

- Integrated KenyaEMR, DHIS2, and AIMS systems using Flask/FastAPI APIs, automating 80% of reporting workflows
- Designed optimized PostgreSQL schemas across 15+ healthcare facilities, improving data quality and consistency
- Delivered comprehensive training to 200+ healthcare staff on analytics, digital systems, and data utilization
- Built Python-based automation tools for QA audits and data validation pipelines

Data Lead

CIHEB Kenya | Jul 2024 - Feb 2025 | Nairobi, Kenya

- Administered PostgreSQL and Oracle databases serving national-level clinical datasets with 99.8% uptime
- Developed Power BI dashboards supporting program monitoring, enabling data-driven decision making
- Strengthened API security implementing JWT/OAuth authentication and authorization protocols
- Led quarterly system audits and established comprehensive documentation standards

Python Developer

Jay & Oak Ltd | Jul 2023 - Jun 2024 | Nairobi, Kenya

- Developed Django REST and Flask APIs with comprehensive data validation and integration logic
- Built responsive web applications consuming APIs for real-time data synchronization
- Implemented automated deployments using Docker containers and Git CI/CD workflows

- Optimized ETL architecture reducing data processing time by 50%

Systems Analyst

Ornac Ltd | Jan 2022 - Jun 2023 | Nairobi, Kenya

- Designed multi-database integration solutions across MySQL, SQL Server, and PostgreSQL environments
- Delivered automation tools for SMEs, reducing manual reporting by 75% and improving workflow efficiency
- Developed text-matching and record-linkage algorithms improving match accuracy by 40%

Research Data Analyst

Aga Khan University Hospital | Jul 2016 - Dec 2021 | Nairobi, Kenya

- Built ETL pipelines in Python to process 100K+ laboratory records, cutting turnaround time by 80%
- Integrated lab APIs with clinical systems ensuring regulatory compliance and data integrity
- Produced automated statistical reports and reproducible analytics scripts using Python and R
- Supported 200+ clinical users in data operations and system utilization

KEY PROJECTS

Mama Africa - Maternity support application for African mothers built with Flask and React, featuring health tracking, appointment scheduling, and community support features.

PeriodTribe - Privacy-first menstrual health tracking app with ML-powered cycle predictions, symptom tracking, and personalized insights.

Vendor Portal (CIHEB Kenya) - Comprehensive vendor prequalification and procurement management system with Django backend and React frontend.

Clinical NLP Response Predictor - Fine-tuned BERT model predicting clinician responses using ROUGE scoring and transfer learning techniques.

Chronicles of the Gods - African mythology-inspired RPG built with Unity 3D featuring combat systems, quest mechanics, and interactive dialogue trees.

EDUCATION

MSc Data Science	University of East London	2025 - 2027
MBA Strategic Management	Liverpool John Moores University	2024 - 2026
BSc Health Sciences	Mount Kenya University	2015

CERTIFICATIONS & TRAINING

AWS Certified Cloud Practitioner (2025) | **Full-Stack Development** - Moringa School (2023) | **Data Science Certification** - Moringa School (2021) | **Kaggle BIPOC Grant Program** (2022-2023) | **IGDA Foundation Scholarship** - Game Development (2023)